Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	295	**2**	**M1** for [87 +] 4×52 oe
1(a)(ii)	29.5 or 29.49…	**1**	**FT** $\frac{87}{\textit{their}\textbf{(a)(i)}} \times 100$
1(b)	11	**2**	**M1** for 18×4 [± 61] oe
1(c)	4160 cao nfww	**2**	**M1** for 64 ÷ 0.0154 or **B1** for rounding *their* answer to nearest 10
1(d)	2.4[0] nfww	**2**	**M1** for $\left(1+\frac{12.5}{100}\right)x = 2.7[0]$ oe
1(e)	53 : 36	**3**	**M2** for 265 : 180 oe or for answer 36 : 53 or 53 min: 36 min or **M1** for 4h 25 [mins] or 265 [mins] seen
1(f)	6[.00] or 5.999…	**3**	**M2** for $\sqrt[5]{\frac{736}{550}}$ or **M1** for $736 = 550 \times (x)^5$
2(a)(i)	3 2.25 1	**3**	**B1** for each
2(a)(ii)	Fully correct smooth curve	**4**	**B3FT** for 7 or 6 correct plots **B2FT** for 5 or 4 correct plots **B1FT** for 3 correct plots
2(a)(iii)	–0.6 to –0.51, 0.75 to 0.85, 1.7 to 1.85	**3**	**B1** for each If 0 scored, **SC1** for $y = 1.5$ drawn
2(a)(iv)	–3 or –2 or –1 or 0	**1**	
2(b)(i)	Tangent ruled at $x = 1$	**1**	
2(b)(ii)	4.4 to 5.6	**2**	Dep on tangent at $x = 1$ or close attempt **M1** for rise/run for *their* line

Question	Answer	Marks	Partial Marks
2(b)(iii)	y = (4.4 to 5.6)x – (1.8 to 2.2) or [y =] *their* **(b)(ii)**x + *their*(y-intercept)	**2**	**FT** for any line but not horizontal or vertical line for 2 marks or **B1** **B1FT** for [m =] *their* 5 or for *their* y-intercept
3(a)	187	**2**	**M1** for $220 \times \left(1 - \frac{15}{100}\right)$ oe or **B1** for 33 seen
3(b)	19.8	**3**	**M2** for $29.7 \times \sqrt[3]{\frac{0.4}{1.35}}$ oe or **M1** for $\sqrt[3]{\frac{0.4}{1.35}}$ or $\sqrt[3]{\frac{1.35}{0.4}}$ oe seen or for $\frac{29.7^3}{x^3} = \frac{1.35}{0.4}$ oe
3(c)	12.4 or 12.44…	**3**	**M1** for $90 \times 75 \times h = 7 \times$ figs 12 **B1** for 1000 cm^3 = 1 litre soi
4(a)	32.9 or 32.91 to 32.92…	**2**	**M1** for $\pi \times 1.65 \times 4.7 + \pi \times 1.65^2$
4(b)	69.4 or 69.44 to 69.45	**2**	**M1** for cos = 1.65 ÷ 4.7 oe
4(c)(i)	12.5 or 12.54 to 12.55	**4**	**M3** for $\frac{1}{3} \times \pi \times 1.65^2 \times \sqrt{4.7^2 - 1.65^2}$ oe or **M2** for $\sqrt{4.7^2 - 1.65^2}$ oe or for $4.7 \times \sin($*their* **(b)**$)$ oe or **M1** for $1.65^2 + h^2 = 4.7^2$ oe or for $\frac{h}{4.7} = \sin($*their* **(b)**$)$ oe
4(c)(ii)	41 nfww	**4**	**B3** for 41.7… to 41.9 or **M2** for $\frac{4}{3} \times \pi \times 5^3 \div$ *their* 12.5 or **M1** for $\frac{4}{3} \times \pi \times 5^3$ After **M2** scored, **M1** for truncating *their* decimal number of cones seen to an integer answer
5(a)	$\frac{10x}{(x-3)(x+2)}$ or $\frac{10x}{x^2 - x - 6}$ final answer	**4**	**M1** for common denominator $(x-3)(x+2)$ isw **M1** for $(x+3)(x+2) - (x-2)(x-3)$ isw **B1** for correct numerator in terms of x only